

important factors include:

- Cost and paid subscriptions
- Censorship and data leakage issues
- Data mismanagement, distribution or selling to unauthorised organisations
- Security and privacy with encryption
- Longer lifetime
- Fault-tolerant
- Robustness in case of node failures

There is a need to integrate secure systems for storage with a better degree of privacy so that any sniffing or cracking may be prevented.

Figure 2 depicts the percentage of data breaches in the health sector in the US that are still under investigation.

Figure 3 shows the huge cost to organisations that have data breaches, as these also affect the integrity of the enterprise.

Assorted platforms for secured backup and recovery

A variety of network based systems, including cloud repositories, are available for uploading files and information, such as Google Drive, DropBox, Microsoft OneDrive, Jio Cloud, and many more. Storage repositories are divided into two categories — controlled cloud storage and decentralised cloud platforms. Anonymity is preserved in the case of decentralised storage platforms through encryption and dynamic fragmentation of data files.

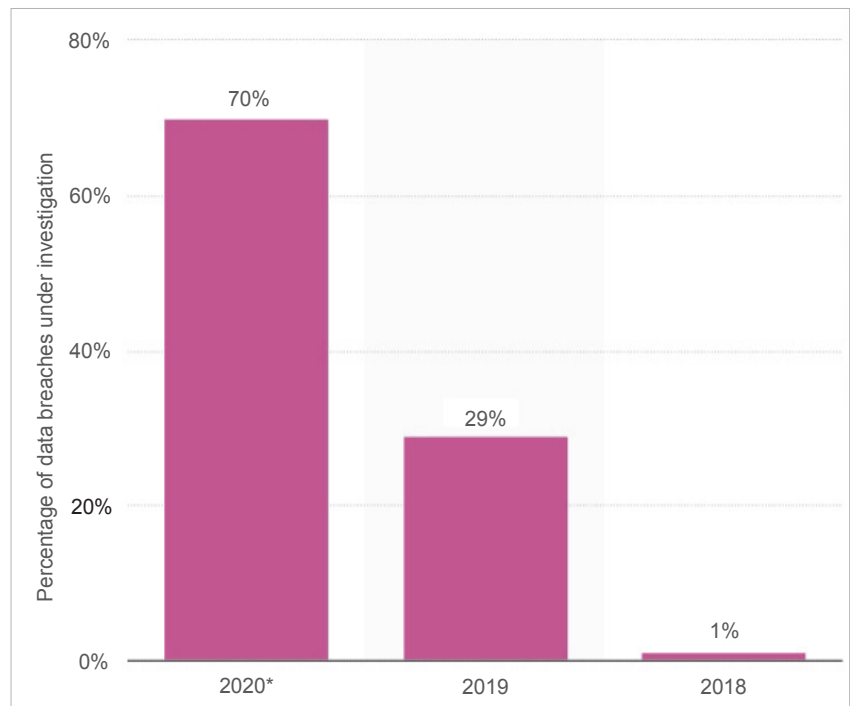


Figure 2: Percentage of data breaches under investigation in the health sector in the US (Source: Research reports of *Statista.com*)

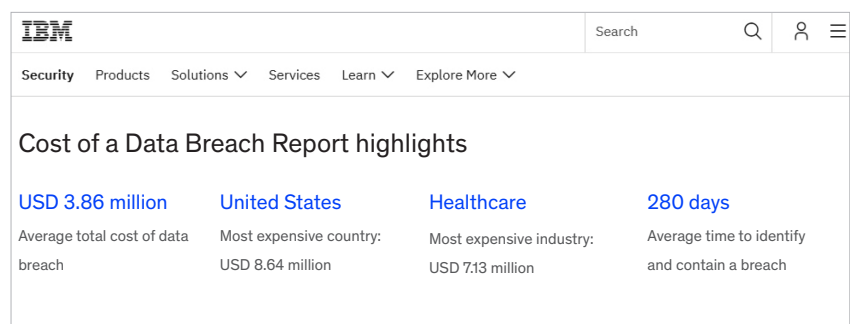


Figure 3: Cost associated with data breaches (Source: IBM)

Cloud platform	URL
IceDrive	https://icedrive.net/
pCloud	https://www.pcloud.com/
Sync	https://www.sync.com/
Mega	https://mega.io/
DropBox	https://www.dropbox.com/
IDrive	https://www.idrive.com/
Yandex	https://disk.yandex.com/
JumpShare	https://jumpshare.com/

Table 1: Centralised platforms

The InterPlanetary File System (IPFS) is one of the decentralised platforms that maintains the security, privacy and integrity of data. Such platforms integrate blockchain technologies to ensure dynamic security for the data files.

Currently, blockchain based cloud storage is used for multiple domains including education, infrastructure, mobile networks, community networks, Web hosting, supply chain management, music, medical data, electronic health records (EHR), and video streaming.

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